Kiki or Bouba? — quick summary

Alper & Averbuch-Elor, NeurIPS 2023

1. The paper in 30 seconds

Question	Do AI models trained on images + captions show the kiki–bouba effect?
Models	CLIP (understands text and images) and Stable Diffusion (draws images from text)
Trick	Invent nonsense words from "sharp" letters (*kitaki*) or "round" letters (*bodubo*)
Test	Check if the models place sharp words near *spiky* shapes and round words near *round* shapes
Answer	Yes, for both models, and humans agree with the models
Why it matters	Sound and meaning aren't fully arbitrary, and models learn this from ordinary captions without hearing any sound

2. Problem and claim

Background

- **Saussure:** a word's sound has nothing to do with its meaning (*dog* vs *chien*).
- **Kiki–bouba effect:** shown a spiky shape and a round shape, almost everyone calls the spiky one *kiki* and the round one *bouba*. True across languages, cultures, even babies.
- So some sounds *do* carry meaning. This is called **sound symbolism**.

Claim

Vision-and-language models have learned sound symbolism: letters for sharp sounds → sharp shapes, letters for round sounds → round shapes.

Assumptions

1. Sounds split into two teams (from phonetics research):
   - **Sharp:** voiceless consonants p t k s h x, vowels e i
   - **Round:** voiced consonants b d g m n l, vowels o u
   - **Neutral:** a
2. Written letters are a good-enough stand-in for sounds (models only see text).
3. CLIP space has a "sharp ↔ round" direction that adjectives can locate.
4. No training at all (zero-shot), so anything found was already in the model.

3. What they built (step by step)

Running example: **kitaki** (sharp) vs **bodubo** (round).

Step 1 — Make nonsense words

- Pattern: 3 syllables, first = last, all letters from one team.
- ki + ta + ki = **kitaki** · bo + du + bo = **bodubo**
- Mixing teams not allowed (*kiduki ✗*).
- Total: **648 words** (324 sharp, 324 round) + the originals *kiki* and *bouba*.

Step 2 — Put them in a prompt

- "a 3D rendering of a **kitaki** shaped object"
- Adjectives too: "a 3D rendering of a **jagged** object"

Step 3 — Turn each prompt into a vector

- **CLIP:** read the prompt text → 1 vector.
- **Stable Diffusion:** draw 50 images → CLIP reads each image → average → 1 vector.
- Both end up in the same CLIP space, so the same test works for both.

Step 4 — Build a "sharpness ruler"

- Add up 10 sharp adjectives (*spiky, jagged, pointed…*), subtract 10 round ones (*round, soft, plump…*).
- What's left is an arrow pointing from "round" to "sharp." Shared parts like "3D object" cancel out.

Step 5 — Score each word on the ruler

Score	Ruler built from	Measures	Example (made-up numbers)
γ (geometric)	adjectives (meaning)	Does a pseudoword look sharp or round?	*kitaki* = +0.50, *bodubo* = −0.40
ϕ (phonetic)	pseudowords (sound)	Does a real word sound sharp or round?	*jagged* = +0.31, *plump* = −0.25

Remember: γ = meaning ruler sorts sounds; ϕ = sound ruler sorts meanings.

4. How they measured it

Test	Question it answers	Simple example
AUC	Pick one sharp and one round word at random: how often does the sharp one score higher?	Sharp wins 6 of 9 pairs → 0.67. (0.5 = guessing, 1.0 = perfect)
Kendall's τ	Do the scores rank in the same order as the true labels?	0 = no relation, 1 = perfect
ΔP_kb	How far apart are *kiki* and *bouba* among all 648 scores?	kiki at 90th percentile, bouba at 10th → 80%
Letter ranking	Which individual letters push toward sharp or round?	Average γ of all words starting with *k*
Real-word ranking	Does the sound ruler (ϕ) sort real English words by shape?	Rank ~6,500 nouns and adjectives; look at both ends
Pictures	Do the drawings actually look sharp or round?	View images for top and bottom words
Human study	Can people tell which word made which image?	Two images + "A = kiki, B = bouba" or the reverse
Corner count	Do sharp-word images have more corners? (doesn't use CLIP)	Harris corner detector on every image

5. Results

Main numbers

	γ AUC	ΔP_kb	ϕ AUC
Stable Diffusion	0.74	80%	0.97
CLIP	0.77	52%	0.98
Random guessing	0.50	0%	0.50

What each result means

1. **Pseudowords sorted correctly** about 3 times out of 4 (γ AUC 0.74–0.77).
2. **Real adjectives sorted almost perfectly** by a ruler built only from nonsense-word letters (ϕ AUC 0.97–0.98).
3. **kiki → sharp, bouba → round** in both models; strongest in Stable Diffusion (80%).
4. **Letters line up like phonetics predicts.** Stable Diffusion, sharp → round:
   - consonants: x h t k p s | g d l n m b
   - vowels: i e a o u
5. **Real words sort by shape:**
   - sharpest: *shard, kite, origami, hexagon, diamond, jagged, angular*
   - roundest: *butterball, yolk, turnip, chubby, plump, squishy*
6. **Pictures match:** *kataka, pitapi* → pointy objects; *bomabo, lobalo* → blobs. Some evoke real things (*donudo* → doughnut, *banaba* → banana).
7. **Humans agree with the model:**
   - kiki vs bouba images: 73% correct (89% after controlling for people and questions)
   - random pseudoword images: 55% correct (78% after controlling)
8. **Corners confirm it:** 12.75 corners per sharp-word image vs 11.47 per round-word image ($p < 10^{-32}$).

Bottom line: both models encode the kiki–bouba pattern, and it matches what people perceive.

6. Checks and limitations

Robustness checks

Worry	Check	Outcome
Only these two models?	DeepFloyd-IF, Kandinsky, GALIP (a GAN)	All show it
Only LAION data?	GALIP trained on different data	Still shows it
Only this prompt?	9 different prompts	All show it
Only English?	Finnish, Indonesian, Hungarian, Lithuanian	All show it
Memorized the experiment?	Only 12 of ~2B training captions mention it; prompting the name gives nonsense images	No
Does vision matter?	Text-only models	Weaker, mixed effect, so vision seems to help

Limitations

- Shows **that** models encode sound symbolism, not **how** or **why**.
- Can't tell if models learned **letter–shape** links or anything about real **sound**.
- English spelling is only a rough guide to sound; two-team split is a simplification.
- ΔP_kb rests on one word pair and flips for some prompts; AUC is the stronger evidence.
- CLIP is both a model being tested and the judge for most scores (corner count is the exception).